



PORTFOLIO CASE STUDY / AUGUST 2026

Elara.ai

Evidence-first automated verification with reproducible citations, deterministic scoring, and auditable AI workflows.

CONTRIBUTION

Sole contributor

TIMELINE

June 2026 - August 2026

STATUS

Feature-complete and owner-validated for the hosted-demo scope

AGENTIC HYBRID RAG

01

Web discovery

Brave search + controlled retrieval

02

Hybrid passage retrieval

Lexical + vector + metadata + provenance

03

Evidence-grounded generation

Approved passages + deterministic scores

DETERMINISTIC SCORING + CITATION GATE

Evidence first, by design.

Elara evaluates one submitted item against timestamped evidence. It preserves the exact sources, passages, snapshots, calculations, versions, and provenance required to reproduce a report.

NOT A LIE DETECTOR

NO PERMANENT CREDIBILITY SCORES

EVIDENCE BOUNDED

13

controlled workflow stages

462

named tests

263

application and evaluation
source files

76

development commits

Deployment: Vercel frontend + owner-controlled AWS demo stack

Core rule: no durable citation audit = no completed report

AGENTIC HYBRID RAG

**Web discovery and hybrid
passage ranking feed
evidence-grounded generation
inside deterministic scoring and
citation gates.**

Who it serves

- Researchers and analysts
- Journalists and policy reviewers
- Technical evaluators who need inspectable evidence

The product problem

Fluency is not evidence

- A polished answer can conceal weak, missing, or circular evidence.
- Repeated reporting can look like independent confirmation.
- Quotes, numbers, attribution, and context need different checks.
- Public pages are untrusted inputs and can attack retrieval systems.
- Missing evidence must remain different from evidence of falsehood.

Success is inspectable

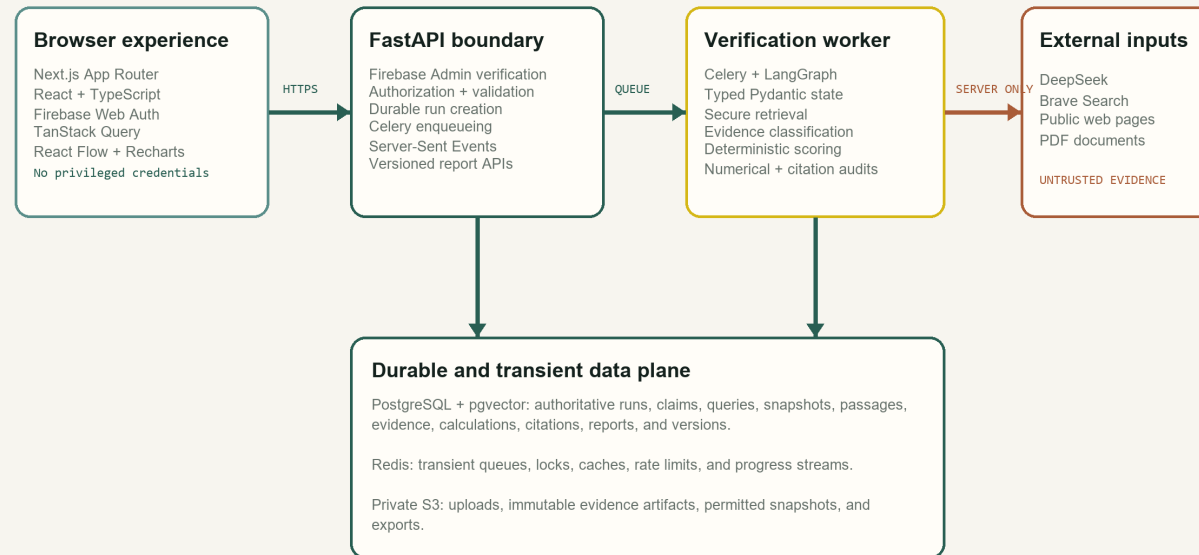
- Every conclusion resolves to an atomic claim and accepted evidence.
- Every factual sentence resolves to an exact stored passage.
- Scoring and arithmetic are reproducible outside the model.
- Contradiction, limitations, and inaccessible sources stay visible.
- Incomplete citation coverage prevents publication.

One item in. Timestamped evidence, explicit uncertainty, and a reproducible assessment out.

Architecture with a hard publication gate

Elara.ai system architecture

A controlled path from user submission to durable, citation-audited report



Publication rule: no report reaches **COMPLETED** until durable citation audit and typed completion gates succeed.

DESIGN DECISIONS

FastAPI is privileged

Auth, authorization, durable run creation, exports, and SSE.

PostgreSQL is truth

Runs, passages, evidence, calculations, reports, and citations persist durably.

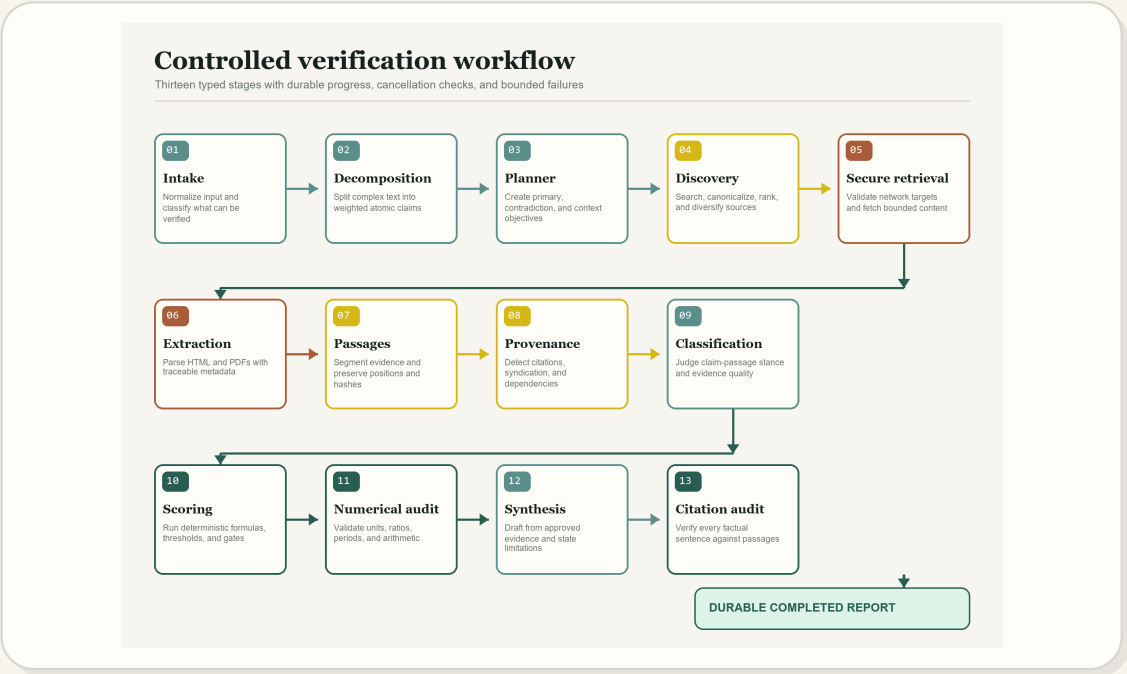
Redis is transient

Queues, locks, rate limits, caches, and progress streams can expire.

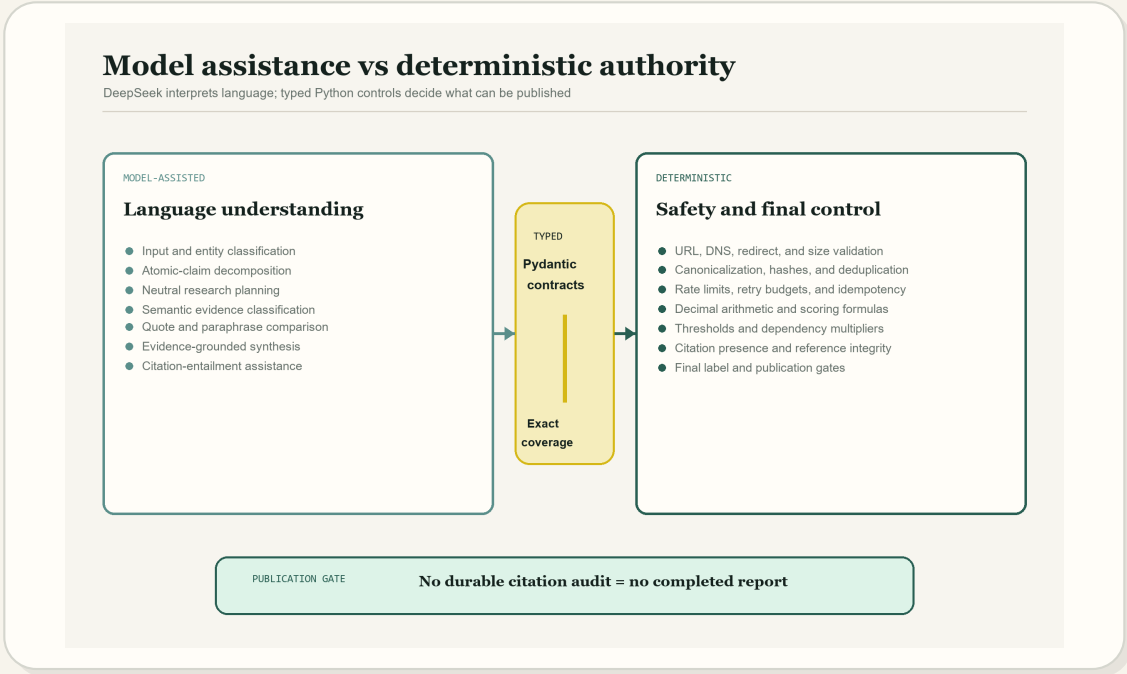
The browser presents

Final scores and report eligibility are never recomputed client-side.

Language understanding, deterministic authority



13 TYPED STAGES



MODEL / DETERMINISTIC BOUNDARY

A fluent draft is never the completion signal.

Durable artifacts, typed state, exact coverage, and citation audit decide what can be published.

One claim, traced end to end

PRIVACY-SAFE REPRESENTATIVE DATA

A proposed transit budget increases funding, adds frequent weekend rail service, and relies primarily on sales-tax revenue.

01 Decompose

Funding, service frequency, and revenue source become separate claims.

02 Retrieve

Budget pages 14 and 22 plus an independent board record are preserved.

03 Compare

The budget supports; the board record exposes pending approval.

04 Calculate

Stored weights and deterministic gates preserve the limitation.

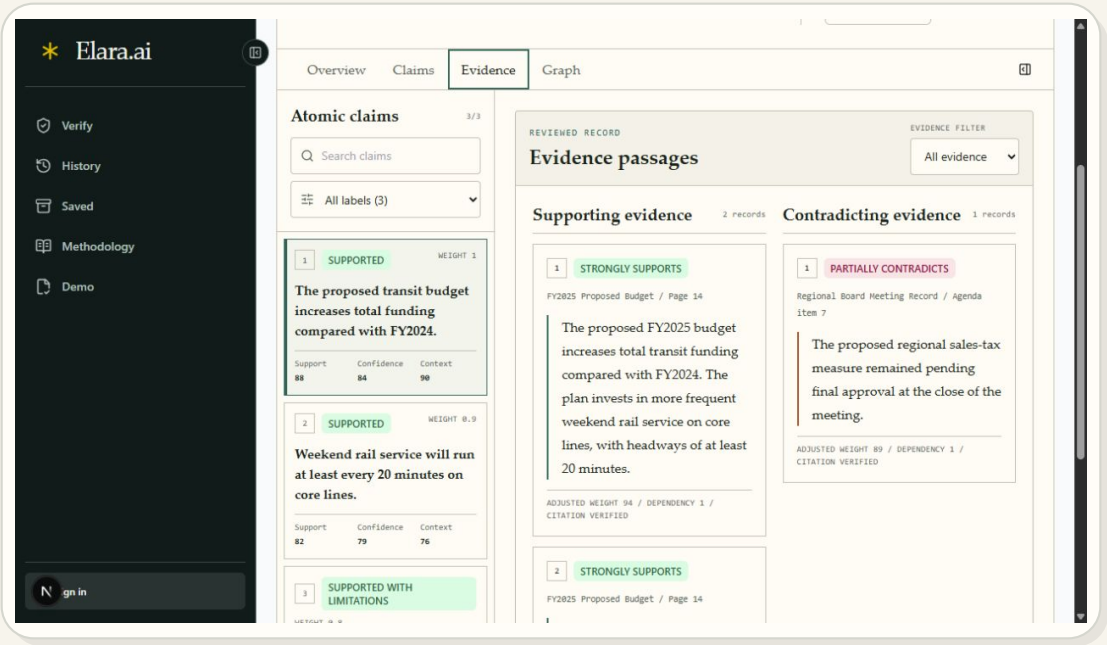
05 Audit

Four factual sentences resolve to stored passages before completion.

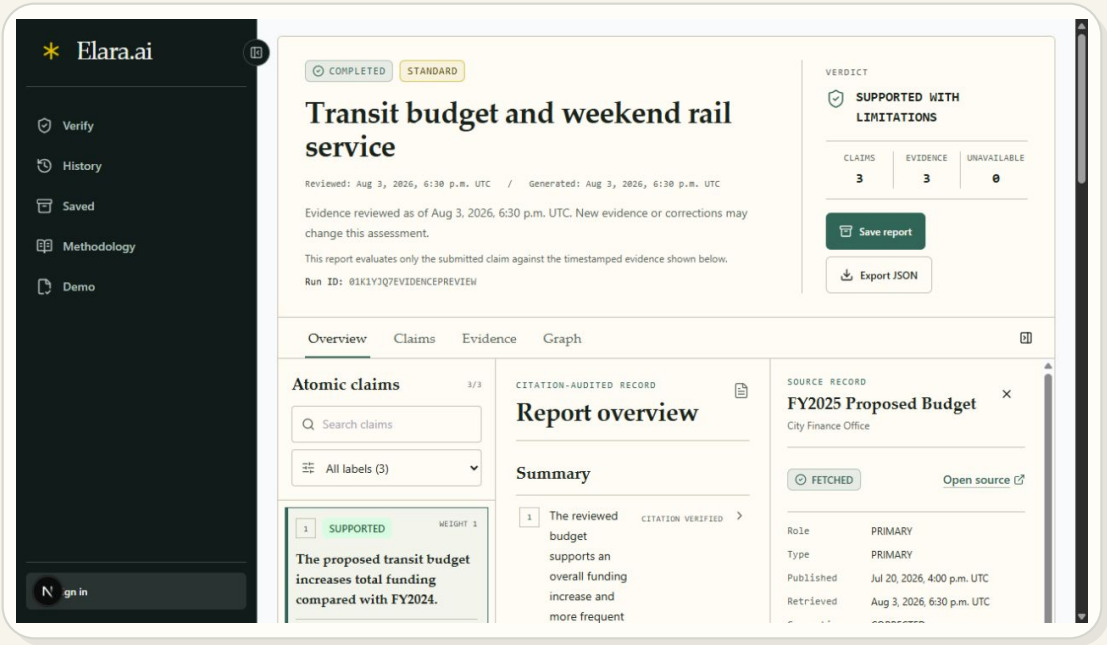
The screenshot displays the Elara.ai interface. On the left is a dark sidebar with navigation links: Verify, History, Saved, Methodology, and Demo. The main content area shows a report titled "Transit budget and weekend rail service" with a status of "COMPLETED" and "STANDARD". It includes a verdict section indicating "SUPPORTED WITH LIMITATIONS" with 3 claims, 3 pieces of evidence, and 0 unavailable items. Below this is a "Report overview" section with a "Summary" and "Factual findings". The "Atomic claims" section shows a claim with a "SUPPORTED" status and a weight of 1. The "Factual findings" section lists two sentences, each with a "CITATION VERIFIED" status.

UI contract preview - placeholder domains, no private evidence

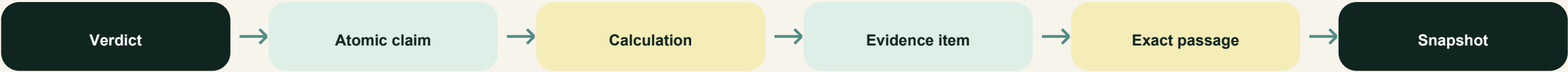
The evidence stays inspectable



SUPPORT AND CONTRADICTION TOGETHER



REPORT SENTENCE TO EXACT PASSAGE



A worked score, outside the model

REPRESENTATIVE FUNDING CLAIM

Supporting weight $P = 66$

Contradicting weight $N = 34$

evidence support = $100 \times P / (P + N)$

66% direction of accepted evidence

Final label: Supported with limitations

The pending approval record and context gates remain visible. A numeric direction never erases a material limitation.



Dependency multipliers discount repeated reporting before evidence contributes to P or N.

Baseline, implementation, and proof are separated

MEASURED BASELINE

Where the time went

218.9 s classification

193.1 s citation audit

Scoring was 0.09 s; numerical audit was 0.06 s. The bottleneck was model-backed language work, not arithmetic.

IMPLEMENTED REMEDIATION

Smaller failure domains

- Classification batch: 4 -> 2 -> 1 task
- Citation batch: 4 -> 2 pairs
- Concurrency capped at 2 model calls
- Schema attempts: 3 -> 2 per batch
- Exact merge coverage; no partial publish

POST-FIX EVIDENCE

Target is not a result

<90 s classification median

<60 s citation-audit median

NOT YET MEASURED

The repository has no completed three-run controlled-live median after the fix. These are acceptance targets, not claimed fixed times.

Reliability and cost outcomes

STRUCTURED OUTPUT

HTTP 200 + invalid schema now produces typed, privacy-safe failure data and bounded local recovery.

OBSERVABILITY

Durable metadata records wall latency, requests, batches, repairs, tokens, model, and prompt version.

SEARCH ENVELOPE

First-phase query targets changed 24/60/120 -> 8/18/36; mandatory coverage still opens phase two.

Problems found, fixes applied, proof retained

Transport tests failed too early

- PROBLEM** Redis, Celery, and S3 security tests stopped at release-revision validation.
- FIX** Built a valid production baseline, then varied only the transport under test.
- PROOF** Intended validators reached; release revision remains independently covered.

Migrations depended on CWD

- PROBLEM** The schema gate could not find migrations when launched from the repository root.
- FIX** Resolved Alembic paths from apps/api/alembic.ini instead of process CWD.
- PROOF** Single-head and upgrade checks pass from root and API contexts.

Legal hold returned the wrong error

- PROBLEM** A held report failed as storage unavailable instead of an explicit conflict.
- FIX** Normalized UTC-aware times and moved the hold check before mutation or cleanup.
- PROOF** Active hold returns 409; records and export objects remain untouched.

Legacy visibility looked permissive

- PROBLEM** A stale test treated visibility=public as cross-user authorization.
- FIX** Kept access owner- or recipient-specific with scope, expiry, and revocation checks.
- PROOF** Unshared, expired, revoked, or wrong-scope access returns non-disclosing 404.

Local storage assumptions leaked

- PROBLEM** MinIO defaults did not match private AWS S3 endpoint and addressing behavior.
- FIX** Separated endpoint, TLS, internal discovery, and path-style configuration.
- PROOF** Regional AWS S3 uses secure transport with forced path style disabled.

Retryable fetch escaped retry policy

- PROBLEM** A source timeout crossed the generic worker boundary and stopped the run.
- FIX** Mapped only retryable FetchError values to the typed fetch failure boundary.
- PROOF** Focused retrieval/task regression suite passed: 37 tests.

Trust boundaries are product features

Server-side secrets

DeepSeek, Brave, Firebase Admin, database, Redis, S3, Sentry auth, and tracing credentials never enter the browser.

Untrusted evidence

Retrieved pages cannot alter workflow instructions, credentials, scoring formulas, or final verdict rules.

Network safety

HTTP(S) only; private, reserved, link-local, metadata, redirect, DNS, port, size, type, and time checks.

Authorization

Runs, sources, snapshots, reports, exports, shares, saved items, and feedback are owner- or recipient-authorized.

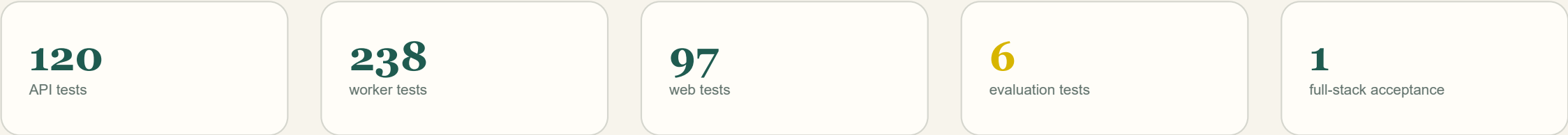
Durable truth

PostgreSQL remains authoritative when Redis locks, queues, cache entries, or progress streams expire.

Fail-closed reports

Missing evidence, invalid structured output, rejected citations, or incomplete audit coverage cannot become complete.

Validation, with the evidence boundary visible



REPRODUCIBLE

Repository gates

Builds, migrations, security regressions, deterministic acceptance, and 462 named tests.

OWNER VALIDATED

Hosted demo

The sole contributor confirmed sign-in, queueing, Celery execution, durable completion, citation-audited retrieval, and refresh.

NOT A BENCHMARK

Methodology evaluation

Offline graders and draft splits exist, but annotations and thresholds remain pending human approval.

No third-party audit or public-production approval is claimed.

Current limitations and deliberate non-goals

What remains bounded

- Human-reviewed evaluation annotations and thresholds are still pending; no independent accuracy benchmark is claimed.
- The transit screenshots use privacy-safe placeholder data, not a live public-authority verification.
- Paywalls, robots restrictions, deleted pages, inaccessible PDFs, and unsupported formats can limit evidence coverage.
- Provider latency and availability vary; the personal demo has no latency or uptime service-level objective.
- Assessments are time-bounded and may change when sources, corrections, or evidence change.
- Security controls are tested, but the project is not independently audited or certified for public launch.

INTENTIONAL DEMO POSTURE

**One EC2 host, manual recovery,
no high availability, no multi-AZ,
no autoscaling, and no enterprise
on-call.**

Simplicity is a scope decision, not an availability claim.

Never a credibility score

Elara evaluates one submitted item against evidence available at a recorded time. It does not score the permanent honesty of a person, organization, or publication.

Built end to end by one contributor

SOLE CONTRIBUTOR

I owned the entire path from product boundary to hosted demonstration.

- Product strategy, UX, design system, and report language
- Next.js frontend, FastAPI API, PostgreSQL model, and migrations
- Celery/LangGraph workflow, DeepSeek and Brave integrations
- Retrieval security, provenance, deterministic scoring, and citation gates
- Tests, evaluation harness, CI, AWS/Vercel deployment, and documentation

Engineering lessons

01

Typed output

HTTP success is not schema success; recovery must preserve exact coverage.

02

Measured stages

Instrumentation showed model calls, not arithmetic, owned the latency.

03

Adaptive cost

Mandatory primary and contradiction paths survive a smaller first-phase budget.

04

Right failure boundary

A fail-closed test matters only when it exercises the boundary named.

Evidence should survive the answer.

Elara shows how language models can operate inside a controlled evidence system without becoming the final authority.

- Model-assisted interpretation; deterministic final control
- Durable evidence, exact passages, calculations, and provenance
- Credible contradiction and explicit limitations
- Citation-gated completion and owner-controlled deployment

Explore the demo

Owner-controlled personal demonstration

[OPEN ELARA.AI DEMO](#)

elara-ai-web.vercel.app

SCOPE NOTE Feature-complete and owner-validated for a low-traffic demo. Human-calibrated methodology evaluation remains future work.